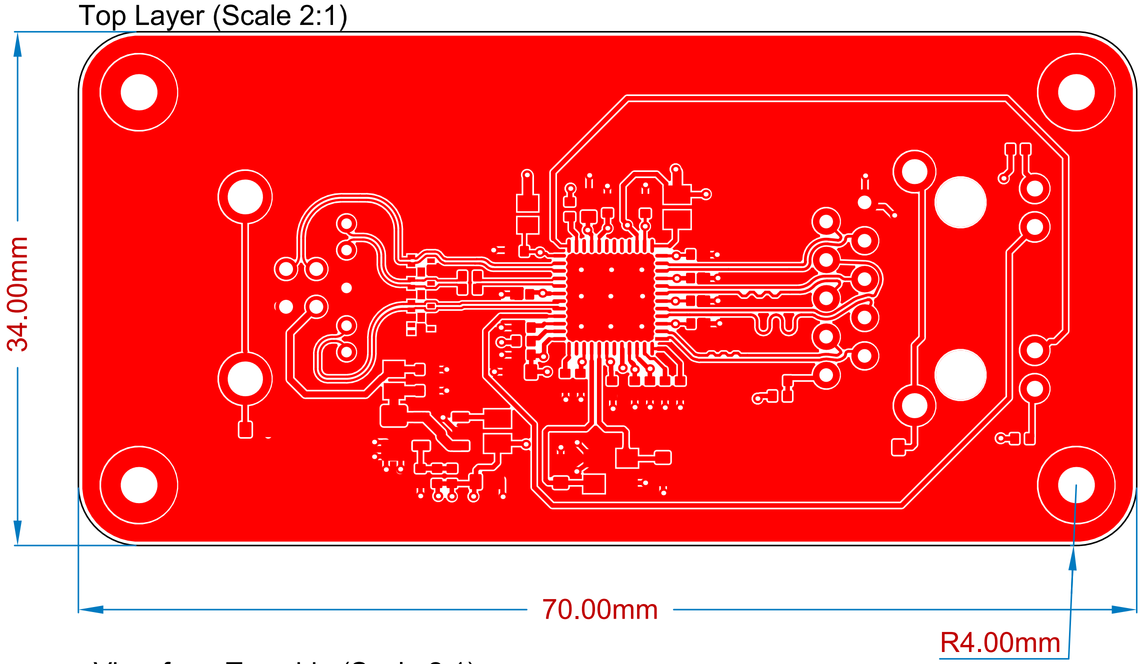
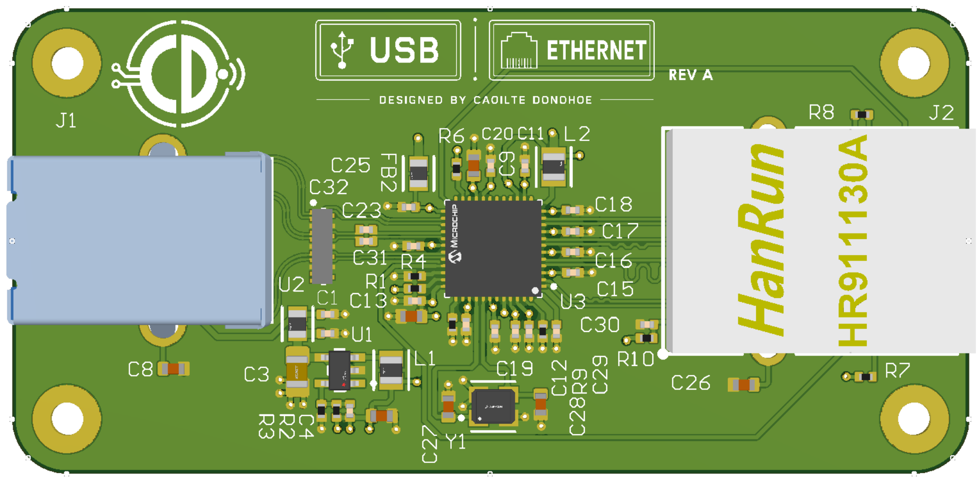
USB 3.1 GEN 1 TO GIGABIT ETHERNET ADAPTER FABRICATION DRAWING

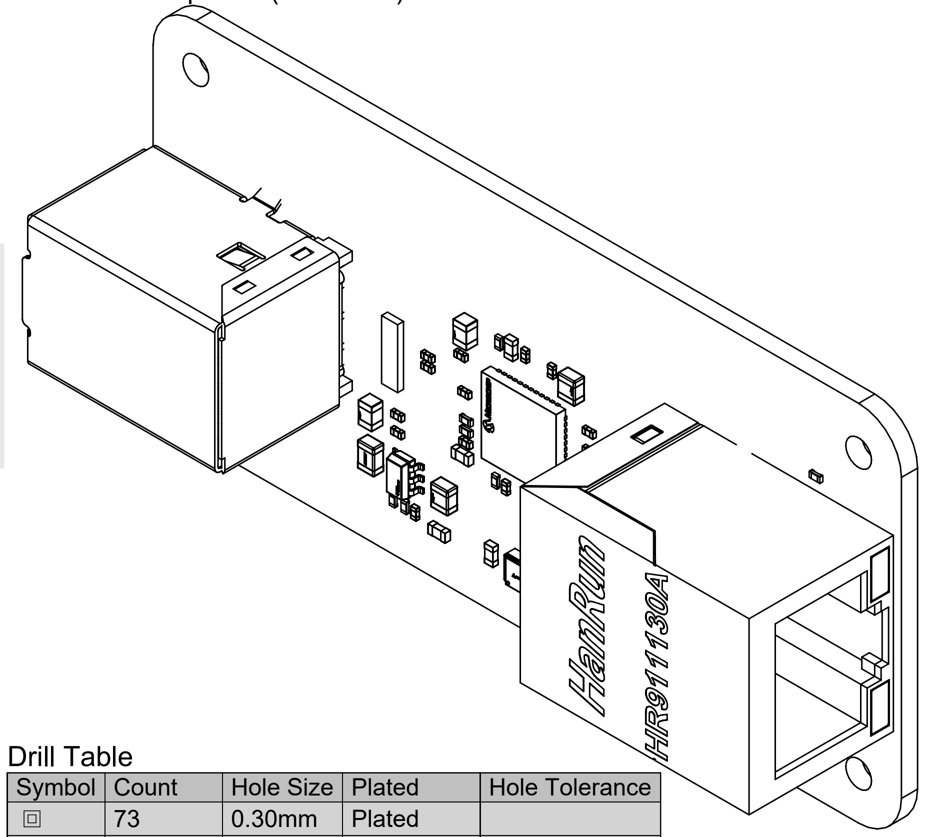
Top Layer (Scale 2:1)



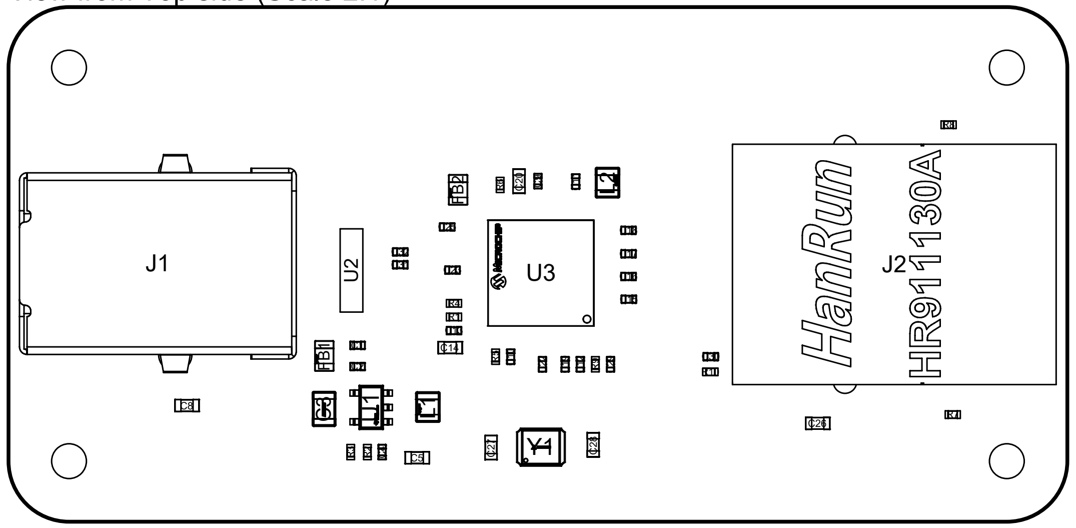
Realistic View: Not to Scale



View from Top side (Scale 2:1)



View from Top side (Scale 2:1)



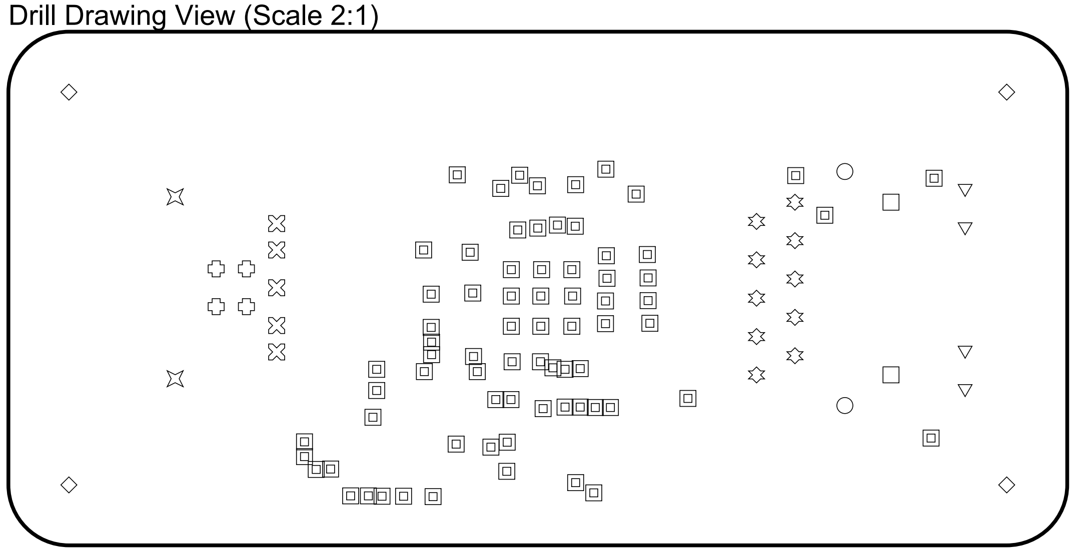
Fabrication Notes:

1. PCB: USB 3.1 Gen 1 to Gigabit Ethernet Adapter
2. 4-layer FR-4 / IS400, finished thickness 1.55 mm
3. Outer / inner copper: 35 µm
4. Inner copper: 35 µm
5. USB2 / USB3 differential impedance: 90 Ω ±10%
6. Gigabit Ethernet differential impedance: 100 Ω ±10%
7. All dimensions are in mm unless otherwise stated.
8. Fabricate to supplied Gerber and NC drill data.
9. Silkscreen: White

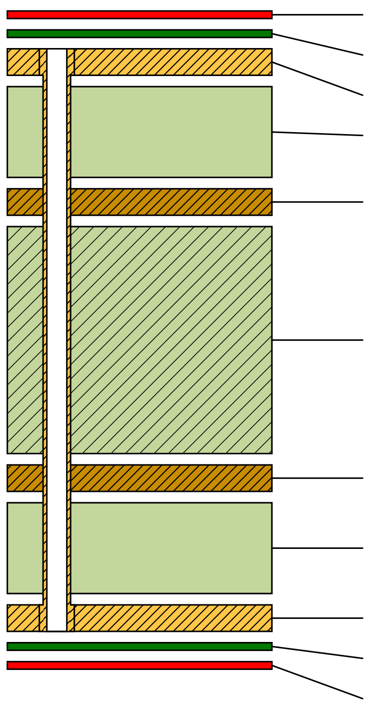
Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
⊡	73	0.30mm	Plated	
⊠	5	0.70mm	Plated	
✩	10	0.90mm	Plated	
⌖	4	0.92mm	Plated	
▽	4	1.05mm	Plated	
○	2	1.65mm	Plated	
⌘	2	2.30mm	Plated	
◇	4	2.40mm	Plated	
□	2	3.25mm	Non-Plated	
	106 Total			

Drill Drawing View (Scale 2:1)



Layer Stack Legend



Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.01mm	Solder Resist	Solder Mask	GTS
Copper	**Top Layer**	**0.04mm**		**Signal**	**GTL**
Prepreg		*0.12mm*	*IS400*	*Dielectric*	
CF-004	**GND_1**	**0.04mm**		**Internal Plane**	**GP1**
		1.13mm	*IS400*	*Dielectric*	
CF-004	**GND_2**	**0.04mm**		**Internal Plane**	**GP2**
Prepreg		*0.12mm*	*IS400*	*Dielectric*	
Copper	**Bottom Layer**	**0.04mm**		**Signal**	**GBL**
Surface Material	Bottom Solder	0.01mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO

Total thickness: 1.53mm

THIRD ANGLE PROJECTION	DRAWN	C. Donohoe	2026-06-01	TITLE:	USB 3.1 Gen 1 to Gigabit Ethernet Adapter
	CHECKED	—	—	SUBTITLE:	Fabrication Drawing
	APPROVED	—	—	PROJECT:	LAN7800 USB Ethernet
				DOCUMENT:	LAN7800_USB_Ethernet_RevA.PCBDwf
UNITS: mm	SCALE: NTS	SHEET: 1 OF 1	REV.: A	DESCRIPTION: Initial Release	DATE: 2026-06-01
DESIGNED BY: Caoilte Donohoe	REV: A				

Design Rules Verification Report

Filename : C:\Users\caoil\Git repos\USB3_Gigabit_Ethernet_Adapter\Hardware\Altium\LA

Warnings 0
Rule Violations 0

Warnings	
Tota	0

Rule Violations	
Clearance Constraint (Gap=0.15mm) (Disabled)(InComponent('U3')),(InComponent('U3'))	0
Clearance Constraint (Gap=1.23mm) (Disabled)(InNetClass('USB3_SS') AND (IsTrack OR IsArc	0
Clearance Constraint (Gap=0.635mm)	0
Clearance Constraint (Gap=0.127mm) (Disabled)(All),(All)	0
Clearance Constraint (Gap=0.1mm) (Disabled)(InComponent('U2')),(InComponent('U2'))	0
Clearance Constraint (Gap=0.1mm) (All),(All)	0
Clearance Constraint (Gap=0.2mm) (Disabled)(InNetClass('USB3_SS') AND	0
Clearance Constraint (Gap=0.08mm) (InNetClass('USB2_HS') OR	0
Short-Circuit Constraint (Allowed=No) (All),(All)	0
Un-Routed Net Constraint ((All))	0
Modified Polygon (Allow modified: No), (Allow shelved: No)	0
Width Constraint (Min=0.15mm) (Max=0.6mm) (Preferred=0.3mm) (InNetClass('POWER_3V3'))	0
Width Constraint (Min=0.4mm) (Max=1mm) (Preferred=0.6mm) (InNetClass('POWER_VBUS'))	0
Width Constraint (Min=0.15mm) (Max=0.3mm) (Preferred=0.2mm)	0
Width Constraint (Min=0.15mm) (Max=1mm) (Preferred=0.2mm) (All)	0
Width Constraint (Min=0.15mm) (Max=1mm) (Preferred=0.2mm) (All)	0
Width Constraint (Min=0.4mm) (Max=1mm) (Preferred=0.6mm) (InNetClass('SWITCH_NODES'))	0
Width Constraint (Min=0.15mm) (Max=0.3mm) (Preferred=0.2mm) (InNetClass('POWER_2V5'))	0
Routing Topology Rule(Topology=Shortest) (All	0
Power Plane Connect Rule(Relief Connect)(Expansion=0.508mm) (Conductor Width=0.254mm)	0
Minimum Annular Ring (Minimum=0.15mm) (All)	0
Hole Size Constraint (Min=0.3mm) (Max=6.35mm) (All)	0
Hole To Hole Clearance (Gap=0.254mm) (All),(All)	0
Minimum Solder Mask Sliver (Gap=0.075mm) (All),(All)	0
Silk To Solder Mask (Clearance=0.1mm) (All),(All)	0
Silk to Silk (Clearance=0.254mm) (All),(All)	0
Net Antennae (Tolerance=0mm) (All)	0
Board Clearance Constraint (Gap=0mm) (All)	0
Matched Lengths(Tolerance=1mm) (InDifferentialPairClass('USB3_DIFF'))	0
Matched Lengths(Tolerance=1.27mm) (InDifferentialPairClass('ETH_DIFF'))	0
Matched Lengths(Tolerance=10mm) (InDifferentialPairClass('ETH_DIFF'))	0
Height Constraint (Min=0mm) (Max=25.4mm) (Prefered=12.7mm) (All)	0
Tota	0